

Lecture 2 – Probability Recap

Last Modified: August 26, 2026

Outline:

2.1	Random Variables	1
2.1.1	Continuous Random Variables	2
2.1.2	Discrete Random Variables	3
2.1.3	Moments and Expectation	3
2.2	Multivariate Random Variables	5
2.3	Properties of Expectation	7
2.4	Transformations of Random Variables	8
2.5	Properties of Special Distributions	12
2.5.1	Normal Distribution	12
2.5.2	Gamma Distribution	12
2.5.3	Binomial Distribution	13
2.5.4	Poisson Distribution	13

2.1 Random Variables

- Each observation X we will collect is a random variable.
- The random sample $\underline{X} = (X_1, \dots, X_n)$ is a vector of independent random variables
- Each statistic $T(\underline{X})$, e.g. $T(\underline{X}) = \sum_{i=1}^n X_i$, is a function of random variables so it is a random variable.
- Let X be a random variable describing a phenomenon of interest. We will always assume that each realization of x is real number. i.e., $x \in \mathbb{R}$. Sometimes we assume $X \in \mathbb{N} = \{1, 2, \dots\}$.
- **Definition:** The **distribution function** of a random variable x is a function F_X such that

$$F_X(x) = \mathbb{P}(X \leq x), \quad \text{for all } x \in \mathbb{R}$$

It is also called the **cumulative distribution function** (cdf).

Example 1. The **Exponential distribution** with parameter $\lambda > 0$ has cdf

$$F_X(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-\lambda x}, & x \geq 0 \end{cases} \quad (2.1)$$

We can write the cdf more compactly using indicator function notation. We write indicators in the following two equivalent ways:

$$\mathbb{1}_{\{x \in A\}} = \mathbb{1}_A(x) = \begin{cases} 0, & x \notin A \\ 1, & x \in A \end{cases}$$

Then the cdf in (2.1) can be expressed compactly as

$$F_X(x) = (1 - e^{-\lambda x}) \mathbb{1}_{\{x \geq 0\}}$$

- If $F_X(x)$ is a continuous function then we say that X is a continuous random variable.
- If $F_X(x)$ is a step function then we say that X is discrete.

2.1.1 Continuous Random Variables

- The **probability density function** (pdf) of a continuous random variable X is defined as

$$f_X(x) = \frac{d}{dx}F_X(x), \quad \forall x \in \mathbb{R}$$

- Knowing the pdf is equivalent to knowing the distribution:

$$F_X(x) = \int_{-\infty}^x f_X(y)dy, \quad \text{and} \quad \mathbb{P}(X \in [a, b]) = \int_a^b f_X(x) dx$$

- Basic properties of a pdf:

$$f(x) \geq 0 \quad \text{and} \quad \int_{-\infty}^{+\infty} f(x)dx = 1$$

- The **Normal distribution** with mean μ and variance σ^2 , denoted $\mathcal{N}(\mu, \sigma^2)$ has density

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

See [here](#) for more on the normal.

- The **Gamma distribution** with shape α and rate β , denoted $\text{Gamma}(\alpha, \beta)$ has density

$$f_X(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x} \mathbb{1}_{\{x \geq 0\}}$$

where $\alpha > 0$, $\beta > 0$, and $\Gamma(\alpha)$ is the Gamma function:

- $\Gamma(1) = 1$, $\Gamma(0) = +\infty$
- $\Gamma(x+1) = x\Gamma(x)$ for all $x \in \mathbb{R}$
- $\Gamma(x) = (x-1)! = 1 \cdot 2 \cdot 3 \cdots (x-1)$ for all $x \in \mathbb{N}$

See [here](#) for more on the gamma distribution.

Example 2. Let $\alpha = 1$. Let us compute the distribution function of $X \sim \text{Gamma}(1, \beta)$. By definition, $F_X(x) = \mathbb{P}(X \leq x)$.

If $x \leq 0$ then

$$\mathbb{P}(X \leq x) = \int_{-\infty}^x f_X(y) dy = \int_{-\infty}^x \beta e^{-\beta y} \mathbb{1}_{\{y \geq 0\}} dy = \int_{-\infty}^x 0 dy = 0.$$

If $x \geq 0$ then

$$\begin{aligned} \mathbb{P}(X \leq x) &= \int_{-\infty}^x f_X(y) dy = \int_{-\infty}^x \beta e^{-\beta y} \mathbb{1}_{\{y \geq 0\}} dy \\ &= \int_0^x \beta e^{-\beta y} dy = -e^{-\beta y} \Big|_0^x = 1 - e^{-\beta x}. \end{aligned}$$

Thus

$$F_X(x) = (1 - e^{-\beta x}) \mathbb{1}_{\{x \geq 0\}}$$

and so $\text{Gamma}(1, \beta)$ is the exponential distribution with rate parameter β .

2.1.2 Discrete Random Variables

- The **probability mass function** (pmf) of a discrete random variable X is

$$f_X(x) = \mathbb{P}(X = x), \quad x = 0, 1, 2, \dots$$

- Knowing the pmf is equivalent to knowing the distribution:

$$F_X(x) = \sum_{y \leq x} f_X(y), \quad \text{and} \quad \mathbb{P}(X \in [a, b]) = \sum_{x: x \in [a, b]} f_X(x)$$

- Basic properties of a pmf

$$f_X(x) \geq 0 \quad \text{and} \quad \sum_{x=0}^{\infty} f_X(x) = 1$$

- The **Bernoulli** distribution with parameter p denoted by $\text{Ber}(p)$ has pmf

$$f_X(x) = \begin{cases} p, & x = 1 \\ 1 - p, & x = 0 \end{cases}$$

- The **Binomial** distribution with parameters n and p denoted by $\text{Bin}(n, p)$ has pmf

$$f_X(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n$$

See [here](#) for more on the binomial.

- The **Poisson** distribution with parameter λ denoted by $\text{Poisson}(\lambda)$ has pmf

$$f_X(x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x = 0, 1, 2, \dots$$

See [here](#) for more on the Poisson.

2.1.3 Moments and Expectation

- The **mean** of a random variable X is

$$\mathbb{E}[X] = \begin{cases} \int_{-\infty}^{\infty} x f_X(x) dx, & \text{continuous case} \\ \sum_{x=0}^{\infty} x f_X(x), & \text{discrete case} \end{cases}$$

It is also called the **expected value** or **first moment**

- The k -th moment of X is given by

$$\mathbb{E}[X^k] = \begin{cases} \int_{-\infty}^{\infty} x^k f_X(x) dx, & \text{continuous case} \\ \sum_{x=0}^{\infty} x^k f_X(x), & \text{discrete case} \end{cases}$$

- Let g be any function. The **expected value** of $g(X)$ is given by

$$\mathbb{E}[g(X)] = \begin{cases} \int_{-\infty}^{\infty} g(x) f_X(x) dx, & \text{continuous case} \\ \sum_{x=0}^{\infty} g(x) f_X(x), & \text{discrete case} \end{cases}$$

Example 3. Let $X \sim \text{Gamma}(\alpha, \beta)$. The expected value is

$$\begin{aligned} \mathbb{E}[X] &= \int_{-\infty}^{\infty} x f(x) dx = \int_0^{\infty} x \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x} dx \\ &= \int_0^{\infty} \frac{\beta^\alpha}{\Gamma(\alpha)} x^\alpha e^{-\beta x} dx = \frac{\beta^\alpha}{\Gamma(\alpha)} \int_0^{\infty} x^\alpha e^{-\beta x} dx. \end{aligned}$$

To solve this integral, we recognize that $x^\alpha e^{-\beta x}$ is the *kernel* of the density of a Gamma distribution with parameters $\alpha + 1$ and β , i.e.,

$$\begin{aligned} \int_0^{\infty} x^\alpha e^{-\beta x} dx &= \int_0^{\infty} x^{(\alpha+1)-1} e^{-\beta x} dx \\ &= \underbrace{\frac{\beta^{\alpha+1}}{\Gamma(\alpha+1)} \frac{\Gamma(\alpha+1)}{\beta^{\alpha+1}}}_{=1} \int_0^{\infty} x^{(\alpha+1)-1} e^{-\beta x} dx \\ &= \frac{\Gamma(\alpha+1)}{\beta^{\alpha+1}} \underbrace{\int_0^{\infty} \frac{\beta^{\alpha+1}}{\Gamma(\alpha+1)} x^{(\alpha+1)-1} e^{-\beta x} dx}_{=1} = \frac{\Gamma(\alpha+1)}{\beta^{\alpha+1}}. \end{aligned}$$

Thus,

$$\mathbb{E}[X] = \frac{\beta^\alpha}{\Gamma(\alpha)} \cdot \frac{\Gamma(\alpha+1)}{\beta^{\alpha+1}} = \frac{\Gamma(\alpha+1)}{\Gamma(\alpha)} \cdot \frac{1}{\beta} = \frac{\alpha}{\beta}.$$

See [here](#) for more on the “massage and squint” approach of recognizing a kernel.

Example 4. $X \sim \text{Poisson}(\lambda)$. The expectation is

$$\mathbb{E}[X] = \sum_{x=0}^{\infty} x f_X(x) = \sum_{x=0}^{\infty} x \frac{\lambda^x e^{-\lambda}}{x!}$$

For $x = 0$ the summand is zero. Thus, we can start the summation from 1, and do the following steps:

$$\begin{aligned} \mathbb{E}[X] &= \sum_{x=1}^{\infty} x \frac{\lambda^x e^{-\lambda}}{x!} = \sum_{x=1}^{\infty} \frac{\lambda^x e^{-\lambda}}{(x-1)!} && \text{using } x/(x!) = 1/(x-1)! \\ &= \lambda \sum_{x=1}^{\infty} \frac{\lambda^{x-1} e^{-\lambda}}{(x-1)!} && \text{pull out one power of } \lambda \\ &= \lambda \underbrace{\sum_{y=0}^{\infty} \frac{\lambda^y e^{-\lambda}}{y!}}_{=1} && \text{reindex: } y = x - 1 \\ &= \lambda. \end{aligned}$$

This is another example of “massage and squint”!

2.2 Multivariate Random Variables

- The random sample $\underline{X} = (X_1, \dots, X_n)$ is a vector of random variables, so we need to work with more than one variable at the same time.
- The following notes consider continuous random variables, but the same ideas extend to the discrete setting by replacing the density function with the mass functions.
- For a pair of random variables (X, Y) the joint distribution is characterized by the joint density function

$$f_{X,Y}(x, y)$$

such that

$$\begin{aligned}\mathbb{P}(X \in [a_1, b_1], Y \in [a_2, b_2]) &= \int_{a_1}^{b_1} \int_{a_2}^{b_2} f_{X,Y}(x, y) dy dx \\ \mathbb{P}(X \leq a, Y \leq b) &= \int_{-\infty}^a \int_{-\infty}^b f_{X,Y}(x, y) dy dx\end{aligned}$$

- The distributions of X and Y (called the **marginal** distributions) can be found as

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy \quad \text{and} \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx \quad (2.2)$$

Let us explain why $f_X(x)$ has to be as in (2.2). The same reasoning applies for $f_Y(y)$. On the one hand,

$$F_X(x) = \mathbb{P}(X \leq x) = \mathbb{P}(X \leq x, Y \leq +\infty) = \int_{-\infty}^x \left(\int_{-\infty}^{+\infty} f_{X,Y}(x, y) dy \right) dx$$

On the other hand, we also know that

$$F_X(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^x f_X(x) dx$$

So matching the integrands recovers the formula: $f_X(x) = \int_{-\infty}^{+\infty} f_{X,Y}(x, y) dy$

Example 5. Consider the joint density

$$f(x, y) = \begin{cases} 6xy^2, & 0 < x < 1 \text{ and } 0 < y < 1 \\ 0, & \text{else} \end{cases}$$

What are the marginal distributions of X and Y ?

For $0 < x < 1$,

$$f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy = \int_0^1 6xy^2 dy = 6x \int_0^1 y^2 dy = 6x \frac{y^3}{3} \Big|_0^1 = 2x$$

Thus $f_X(x) = 2x \mathbb{1}_{\{0 < x < 1\}}$.

For $0 < y < 1$,

$$f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx = \int_0^1 6xy^2 dx = 6y^2 \int_0^1 x dx = 6y^2 \frac{x^2}{2} \Big|_0^1 = 3y^2$$

Thus $f_Y(y) = 3y^2 \mathbb{1}_{\{0 < y < 1\}}$.

- Often we observe one of the variables (e.g., $Y = y$) and want the distribution of X conditional on $Y = y$. Thus, we define the **conditional density function** as

$$f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

- The **conditional expectation** becomes

$$\mathbb{E}[X | Y = y] = \int_{-\infty}^{+\infty} x f_{X|Y}(x | y) dx$$

Note that this is a function of y .

Example 6. Consider the joint density

$$f(x, y) = e^{-y} \mathbb{1}_{\{0 < x < y < +\infty\}}$$

Let us compute $\mathbb{E}[Y | X = x]$.

First we find the marginal distribution of X . For $x \geq 0$,

$$f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy = \int_0^{+\infty} e^{-y} \mathbb{1}_{\{x \leq y\}} dy = \int_x^{+\infty} e^{-y} dy = -e^{-y} \Big|_x^{+\infty} = e^{-x}$$

We recognize that X has an exponential distribution with parameter 1. Thus,

$$f_{Y|X}(y | x) = \frac{f(x, y)}{f_X(x)} = \frac{e^{-y}}{e^{-x}} \mathbb{1}_{\{x < y\}} = e^{-(y-x)} \mathbb{1}_{\{x < y\}}$$

The conditional expectation is then

$$\begin{aligned} \mathbb{E}[Y | X = x] &= \int_{-\infty}^{+\infty} y f_{Y|X}(y | x) dy = \int_x^{+\infty} y e^{-(y-x)} dy \\ &= \int_0^{+\infty} (z + x) e^{-z} dz = \int_0^{+\infty} z e^{-z} dz + x \int_0^{+\infty} e^{-z} dz = 1 + x \end{aligned}$$

where the third equality follows from the substitution $z = y - x$.

Definition 1. Random variables X and Y are **independent** if and only if

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

- In the **multidimensional** case (more than two variables) the distribution of vector $\underline{X} = (X_1, \dots, X_n)$ of random variables is characterized by the **joint density function**

$$f_{\underline{X}}(x_1, \dots, x_n)$$

such that

$$\mathbb{P}(X_1 \in [a_1, b_1], \dots, X_n \in [a_n, b_n]) = \int_{a_1}^{b_1} \cdots \int_{a_n}^{b_n} f_{\underline{X}}(x_1, \dots, x_n) dx_1 \dots dx_n$$

- For a random vector $(\underline{X}, Y) = (X_1, \dots, X_n, Y)$ with joint density $f_{\underline{X},Y}(x, y)$ the conditional density is

$$f_{\underline{X}|Y}(\underline{x} | y) = \frac{f_{\underline{X},Y}(x_1, \dots, x_n, y)}{f_Y(y)}$$

Definition 2. Random variables $X_1, \dots, X_n$ are **independent** if and only if

$$f_{\underline{X}}(x_1, \dots, x_n) = f_{X_1}(x_1) f_{X_2}(x_2) \cdots f_{X_n}(x_n) = \prod_{i=1}^n f_{X_i}(x_i)$$

2.3 Properties of Expectation

- The expected value is **linear**. This means that

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y] \quad \text{and} \quad \mathbb{E}[cX] = c \mathbb{E}[X]$$

for any random variables X and Y and any constant c . Combining these properties yields:

$$\mathbb{E}\left[\sum_{i=1}^n c_i X_i\right] = \sum_{i=1}^n c_i \mathbb{E}[X_i]$$

- If random variables X and Y are **independent** then

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$$

- If $X_1, \dots, X_n$ are independent random variables then

$$\mathbb{E}\left[\prod_{i=1}^n X_i\right] = \prod_{i=1}^n \mathbb{E}[X_i]$$

Definition 3. The **variance** of a random variable is given by

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$$

This is a measure of how much X deviates from its mean $\mathbb{E}[X]$.

Lemma 2.1. We have $\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$

Proof.

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[(X - \mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2 - 2X\mathbb{E}[X] + \mathbb{E}[X]^2] \\ &= \mathbb{E}[X^2] + \mathbb{E}[-2X\mathbb{E}[X]] + \mathbb{E}[\mathbb{E}[X]^2] && \text{linearity of expectation} \\ &= \mathbb{E}[X^2] - 2\mathbb{E}[X]\mathbb{E}[X] + \mathbb{E}[X]^2 && -2\mathbb{E}[X] \text{ and } \mathbb{E}[X]^2 \text{ are constants} \\ &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 \end{aligned}$$

□

Properties of the variance:

- $\text{Var}(X) \geq 0$ and $\text{Var}(X) = 0$ if and only if $X = c$ for a constant c .
- $\text{Var}(cX) = c^2 \text{Var}(X)$.
- If X and Y are independent then $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$.
- If $X_1, \dots, X_n$ are independent then

$$\text{Var}\left(\sum_{i=1}^n c_i X_i\right) = \sum_{i=1}^n c_i^2 \text{Var}(X_i)$$

Example 7. Consider again the joint density

$$f(x, y) = \begin{cases} 6xy^2, & 0 < x < 1 \text{ and } 0 < y < 1 \\ 0, & \text{else} \end{cases}$$

Let us compute $\text{Var}(2X - 3Y)$

- Previously, we computed the marginal densities as

$$f_X(x) = 2x \mathbb{1}_{\{0 < x < 1\}} \quad \text{and} \quad f_Y(y) = 3y^2 \mathbb{1}_{\{0 < y < 1\}}$$

- Observing that

$$f_X(x)f_Y(y) = cxy^2 \mathbb{1}_{\{0 < x < 1\}} \mathbb{1}_{\{0 < y < 1\}}$$

we see that X and Y are independent, and thus

$$\text{Var}(2X - 3Y) = \text{Var}(2X) + \text{Var}(-3Y) = 4 \text{Var}(X) + 9 \text{Var}(Y)$$

- Next we compute the first and second moments:

$$\begin{aligned} \mathbb{E}[X] &= \int_{-\infty}^{+\infty} x f_X(x) dx = \int_0^1 2x^2 dx = 2 \frac{x^3}{3} \Big|_0^1 = \frac{2}{3} \\ \mathbb{E}[X^2] &= \int_{-\infty}^{+\infty} x^2 f_X(x) dx = \int_0^1 2x^3 dx = 2 \frac{x^4}{4} \Big|_0^1 = \frac{1}{2} \\ \mathbb{E}[Y] &= \int_{-\infty}^{+\infty} y f_Y(y) dy = \int_0^1 3y^3 dy = 3 \frac{y^4}{4} \Big|_0^1 = \frac{3}{4} \\ \mathbb{E}[Y^2] &= \int_{-\infty}^{+\infty} y^2 f_Y(y) dy = \int_0^1 3y^4 dy = 3 \frac{y^5}{5} \Big|_0^1 = \frac{3}{5} \end{aligned}$$

- Putting everything back together, we have

$$\text{Var}(X) = \frac{1}{2} - \frac{4}{9} = \frac{1}{18} \quad \text{and} \quad \text{Var}(Y) = \frac{3}{5} - \frac{9}{16} = \frac{3}{80}$$

so

$$\text{Var}(2X - 3Y) = \frac{4}{18} + \frac{27}{80}$$

2.4 Transformations of Random Variables

- Let X be a random variable. Often we are in a function $g(X)$, which is again a random variable.
- Let X have sample space / domain $\mathcal{X}$ (which is either $\mathbb{R}$ or $\mathbb{N}$) and let $g: \mathcal{X} \rightarrow \mathcal{Y}$ be a function (e.g., $g(x) = x^2$). Then $Y = g(X)$ is a random variable with domain $\mathcal{Y}$.
- How do we obtain the distribution of Y ? Observe that for any set $A \subset \mathcal{Y}$,

$$\mathbb{P}(Y \in A) = \mathbb{P}(g(X) \in A) = \mathbb{P}(X \in \{x \in \mathcal{X} \mid g(x) \in A\}) = \mathbb{P}(X \in g^{-1}(A))$$

where $g^{-1}(A) = \{x \in \mathcal{X} \mid g(x) \in A\}$ is the set of values in the domain that map to points in the set A .

- If for every y there exists a single x such that $g(x) = y$, the g is **invertible** and we can define inverse $g^{-1}: \mathcal{Y} \rightarrow \mathcal{X}$.
- **Discrete case:** If $\mathcal{X}$ is finite or countable then

$$\mathcal{Y} = \{y \mid y = g(x) \text{ for some } x \in \mathcal{X}\}$$

is finite or countable and so Y is also a discrete random variable. The probability mass function is

$$f_Y(y) = \mathbb{P}(Y = y) = \sum_{x \in g^{-1}(\{y\})} \mathbb{P}(X = x) = \sum_{x \in g^{-1}(\{y\})} f_X(x)$$

Example 8. If $X \sim \text{Poisson}(\lambda)$ and $Y = 2X$, then

$$\mathcal{X} = \{0, 1, 2, \dots\} \implies \mathcal{Y} = \{0, 2, 4, \dots\}$$

Moreover, if $y \in \mathcal{Y}$ then

$$f_Y(y) = \sum_{x \in g^{-1}(\{y\})} f_X(x) = f_X(y/2) = \frac{\lambda^{y/2} e^{-\lambda}}{(y/2)!}$$

Here, we see that $g(x) = 2x$ is invertible with $g^{-1}(y) = y/2$.

- **Continuous case:** more complicated! Define

$$\mathcal{X} = \{x \mid f_X(x) > 0\} \quad \text{and} \quad \mathcal{Y} = \{y \mid y = g(x) \text{ for some } x \in \mathcal{X}\}$$

Theorem 1 (Change of variable). Let g be strictly monotone (increasing or decreasing) and assume the inverse g^{-1} has a continuous derivative on $\mathcal{Y}$. Then

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

Note: Under the assumptions of the theorem, g is strictly monotone and so the inverse g^{-1} is well-defined.

Proof. First suppose that g is increasing. Then,

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(g(X) \leq y) = \mathbb{P}(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$$

Differentiating both sides gives

$$\begin{aligned} f_Y(y) &= \frac{d}{dy} F_Y(y) = \frac{d}{dy} F_X(g^{-1}(y)) \\ &= f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y) && \text{chain rule} \\ &= f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right| && g \text{ is increasing} \end{aligned}$$

Next suppose that g is decreasing. In this case,

$$\begin{aligned} F_Y(y) &= \mathbb{P}(Y \leq y) = \mathbb{P}(g(X) \leq y) = \mathbb{P}(X \geq g^{-1}(y)) \\ &= 1 - \mathbb{P}(X \leq g^{-1}(y)) = 1 - F_X(g^{-1}(y)) \end{aligned}$$

and so differentiating gives

$$\begin{aligned}
 f_Y(y) &= \frac{d}{dy} F_Y(y) = \frac{d}{dy} [1 - F_X(g^{-1}(y))] \\
 &= -f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y) && \text{chain rule} \\
 &= f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right| && g \text{ is decreasing}
 \end{aligned}$$

□

Example 9. Suppose $X \sim \text{Gamma}(\alpha, \beta)$ and $Y = X^2$.

- $\mathcal{X} = \mathbb{R}_+$ and $g(x) = x^2$ is increasing on $\mathbb{R}_+$ with $g^{-1}(y) = \sqrt{y}$.

- Moreover,

$$\frac{d}{dy} g^{-1}(y) = \frac{d}{dy} \sqrt{y} = \frac{1}{2\sqrt{y}}$$

- Thus,

$$\begin{aligned}
 f_Y(y) &= f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right| \\
 &= \frac{\beta^\alpha}{\Gamma(\alpha)} (\sqrt{y})^{\alpha-1} e^{-\beta\sqrt{y}} \frac{1}{2\sqrt{y}} \mathbb{1}_{\{y>0\}} \\
 &= \frac{\beta^\alpha}{2\Gamma(\alpha)} y^{\frac{\alpha}{2}-1} e^{-\beta\sqrt{y}} \mathbb{1}_{\{y>0\}}
 \end{aligned}$$

- The change of variable formula is not always applicable since g may not be monotone. Another approach is to work with the cumulative distribution function (cdf) $F_Y(y)$ and then differentiate to obtain the density $f_Y(y)$.

Example 10. Let $X \sim \mathcal{N}(0, 1)$. Find the distribution of $Y = X^2$.

- $\mathcal{X} = \mathbb{R}$ and $g(x) = x^2$ is *not* monotone!.
- Let us try to compute $F_Y(y)$ with $y \in \mathcal{Y} = \mathbb{R}_+$.

$$\begin{aligned}
 F_Y(y) &= \mathbb{P}(Y \leq y) = \mathbb{P}(X^2 \leq y) \\
 &= \mathbb{P}(-\sqrt{y} \leq X \leq \sqrt{y}) \\
 &= \mathbb{P}(X \leq \sqrt{y}) - \mathbb{P}(X < -\sqrt{y}) && \text{since } [-\sqrt{y}, \sqrt{y}] = \{x \leq \sqrt{y}\} \setminus \{x \leq -\sqrt{y}\} \\
 &= \mathbb{P}(X \leq \sqrt{y}) - \mathbb{P}(X \leq -\sqrt{y}) && X \text{ is continuous} \\
 &= F_X(\sqrt{y}) - F_X(-\sqrt{y})
 \end{aligned}$$

- Differentiating,

$$\begin{aligned}
 f_Y(y) &= \frac{d}{dy} F_Y(y) = \frac{d}{dy} F_X(\sqrt{y}) - \frac{d}{dy} F_X(-\sqrt{y}) \\
 &= \frac{1}{2\sqrt{y}} f_X(\sqrt{y}) + \frac{1}{2\sqrt{y}} f_X(-\sqrt{y})
 \end{aligned}$$

- Note that this relation holds for any continuous X !

- Recalling that $X \sim \mathcal{N}(0, 1)$, we have $f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ and so

$$\begin{aligned} f_Y(y) &= \frac{1}{2\sqrt{y}} f_X(\sqrt{y}) + \frac{1}{2\sqrt{y}} f_X(-\sqrt{y}) \\ &= \frac{1}{2\sqrt{y}} \frac{1}{\sqrt{2\pi}} e^{-\frac{y}{2}} + \frac{1}{2\sqrt{y}} \frac{1}{\sqrt{2\pi}} e^{-\frac{y}{2}} \\ &= \frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{y}} e^{-\frac{y}{2}} \end{aligned}$$

- Finally, we conclude that

$$f_Y(y) = \frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{y}} e^{-\frac{y}{2}} \mathbb{1}_{\{y \geq 0\}}$$

This is the Gamma distribution with parameters $\alpha = \frac{1}{2}$ and $\beta = \frac{1}{2}$ (note that $\Gamma(1/2) = \sqrt{\pi}$).

- The distribution of Y in the previous example is also called the **chi-squared distribution** with 1 degree of freedom. More generally, the **chi-squared distribution** with k -degrees of freedom corresponds to $\text{Gamma}(\frac{k}{2}, \frac{1}{2})$ and has density

$$f_X(x) = \frac{1}{2^{\frac{n}{2}} \Gamma(\frac{k}{2})} x^{\frac{k}{2}-1} e^{-\frac{x}{2}} \mathbb{1}_{\{x \geq 0\}}$$

It is denoted by χ_k^2 .

- The **Beta distribution** with parameters α and β is denoted by $\text{Beta}(\alpha, \beta)$ and has pdf

$$f_X(x) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1} \mathbb{1}_{\{x \in (0,1)\}}$$

- If $\alpha = \beta = 1$ then the the $\text{Beta}(1, 1)$ distribution has pdf

$$f_X(x) = \mathbb{1}_{\{x \in (0,1)\}}$$

which is called the **uniform distribution** on $(0, 1)$.

- More generally, the **uniform distribution** on interval (a, b) is denoted by $\text{Unif}(a, b)$ and has pdf

$$f_X(x) = \frac{1}{b-a} \mathbb{1}_{\{x \in (a,b)\}}$$

Example 11. Let $X_i \stackrel{\text{iid}}{\sim} \text{Unif}(0, 1)$ for $1 \leq i \leq n$, that is, $X_1, \dots, X_n$ are independent and identically distributed according to $\text{Unif}(0, 1)$. Find the distribution of $Y = \max\{X_1, \dots, X_n\}$.

- Notice that $\mathcal{Y} = (0, 1)$. For $y \in \mathcal{Y}$,

$$\begin{aligned} F_Y(y) &= \mathbb{P}(Y \leq y) \\ &= \mathbb{P}(\max\{X_1, \dots, X_n\} \leq y) \\ &= \mathbb{P}(X_1 \leq y, \dots, X_n \leq y) \\ &= \mathbb{P}(X_1 \leq y) \mathbb{P}(X_2 \leq y) \cdots \mathbb{P}(X_n \leq y) && \text{independent} \\ &= \mathbb{P}(X_1 \leq y)^n && \text{identically distributed} \\ &= (F_{X_1}(y))^n \end{aligned}$$

- Differentiating,

$$\begin{aligned} f_Y(y) &= \frac{d}{dy} f_Y(y) = \frac{d}{dy} (F_{X_1}(y))^n \\ &= n(F_{X_1}(y))^{n-1} \cdot \left(\frac{d}{dy} F_{X_1}(y) \right) \\ &= n(F_{X_1}(y))^{n-1} \cdot f_{X_1}(y). \end{aligned}$$

Note that this relation holds for any continuous distribution of X_1 .

- In our case,

$$\begin{aligned} f_{X_1}(y) &= \mathbb{1}_{\{y \in (0,1)\}} \\ F_{X_1}(y) &= \mathbb{P}(X_1 \leq y) = \int_{-\infty}^y f_{X_1}(x) dx = \int_0^y 1 dx = y \end{aligned}$$

- Thus,

$$\begin{aligned} f_Y(y) &= ny^{n-1} \mathbb{1}_{\{y \in (0,1)\}} = ny^{n-1} (1-y)^{1-1} \mathbb{1}_{\{y \in (0,1)\}} \\ &= \frac{\Gamma(n+1)}{\Gamma(1)\Gamma(n)} y^{n-1} (1-y)^{1-1} \mathbb{1}_{\{y \in (0,1)\}} \end{aligned}$$

and so $Y \sim \text{Beta}(n, 1)$.

2.5 Properties of Special Distributions

2.5.1 Normal Distribution

- Let $X \sim \mathcal{N}(\mu, \sigma^2)$. The expectation and variance of X are

$$\mathbb{E}[X] = \mu, \quad \text{Var}(X) = \sigma^2.$$

The normal distribution is determined completely by its mean and variance.

- Let $X \sim \mathcal{N}(\mu, \sigma^2)$. For every constant c ,
 1. $Y = c + X$ has distribution $\mathcal{N}(\mu + c, \sigma^2)$
 2. $Y = cX$ has distribution $\mathcal{N}(c\mu, c^2\sigma^2)$
- If $Z \sim \mathcal{N}(0, 1)$, then $\mu + \sigma Z \sim \mathcal{N}(\mu, \sigma^2)$. If $X \sim \mathcal{N}(\mu, \sigma^2)$, we can also write $X \stackrel{d}{=} \mu + \sigma Z$, where $\stackrel{d}{=}$ denotes equality in distribution.
- If $X \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $Y \sim \mathcal{N}(\mu_2, \sigma_2^2)$ are **independent** then their sum is normal with matching mean and variance:

$$X + Y \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

2.5.2 Gamma Distribution

- Let $X \sim \text{Gamma}(\alpha, \beta)$. For every constant c , the random variable $Y = cX$ has distribution $\text{Gamma}(\alpha, \frac{\beta}{c})$. Therefore,

$$X \stackrel{d}{=} \frac{1}{\beta} Z \quad \text{where } Z \sim \text{Gamma}(\alpha, 1)$$

- If $X \sim \text{Gamma}(\alpha_1, \beta)$ and $Y \sim \text{Gamma}(\alpha_2, \beta)$ are **independent** then their sum is also gamma:

$$X + Y \sim \text{Gamma}(\alpha_1 + \alpha_2, \beta)$$

2.5.3 Binomial Distribution

- The **Binomial Distribution** with parameters $n \in \mathbb{N}$ and $p \in (0, 1)$ is denoted by $\text{Bin}(n, p)$ and has pmf

$$f_X(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x \in \{0, 1, 2, \dots, n\}$$

- If $X \sim \text{Bin}(n_1, p)$ and $Y \sim \text{Bin}(n_2, p)$ are **independent** then their sum is also binomial

$$X + Y \sim \text{Bin}(n_1 + n_2, p)$$

- If $Z_1, \dots, Z_n \stackrel{\text{iid}}{\sim} \text{Ber}(p)$, then $Z_1 + \dots + Z_n \sim \text{Bin}(n, p)$.

2.5.4 Poisson Distribution

- If $X \sim \text{Poisson}(\lambda_1)$ and $Y \sim \text{Poisson}(\lambda_2)$ are **independent** then their sum is also Poisson

$$X + Y \sim \text{Poisson}(\lambda_1 + \lambda_2)$$